

Jerte Valley cherry-enriched diets improve nocturnal rest and increase 6-sulfatoxymelatonin and total antioxidant capacity in the urine of middle-a...

Tryptophan, serotonin, and melatonin, present in Jerte Valley cherries, participate in sleep regulation and exhibit antioxidant properties. The effect of the intake of seven different Jerte Valley cherry cultivars on the sleep-wake cycle, 6-sulfatoxymelatonin levels, and urinary total antioxidant capacity in middle-aged and elderly participants was evaluated. Volunteers were subjected to actigraphic monitoring to record and display the temporal patterns of their nocturnal activity and rest. 6-sulfatoxymelatonin and total antioxidant capacity were quantified by enzyme-linked immunosorbent assay and colorimetric assay kits, respectively. The intake of each of the cherry cultivars produced beneficial effects on actual sleep time, total nocturnal activity, assumed sleep, and immobility. Also, there were significant increases in 6-sulfatoxymelatonin levels and total antioxidant capacity in urine after the intake of each cultivar. These findings suggested that the intake of Jerte Valley cherries exerted positive effect on sleep and may be seen as a potential nutraceutical tool to counteract oxidation.